NIT2112
Object Oriented Programming

Session 02

Encapsulation and Information Hiding

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Acknowledgement of Country

Victoria University acknowledges, recognises and respects the Ancestors, Elders and families of the Bunurong/Boonwurrung, Wadawurrung and Wurundjeri/Woiwurrung of the Kulin who are the traditional owners of University land in Victoria, the Gadigal and Guring-gai of the Eora Nation who are the traditional owners of University land in Sydney, and the Yugara/YUgarapul people and Turrbal people living in Meanjin (Brisbane).

Session Outline

- Encapsulation: Definition & Benefits
- Access Modifiers (Public, Protected, Private)
- Information Hiding: Getters and Setters
- Name Mangling Explained
- Encapsulation Best Practices

Encapsulation: Bundling and Hiding

- **Encapsulation** is the bundling of **data (attributes)** and **methods** that operate on that data into a single unit (a class). It also involves **hiding** the internal state of an object and requiring all interactions to be performed through the object's methods.

- A capsule containing medicine – you only interact with the capsule as a whole, not the individual ingredients directly.

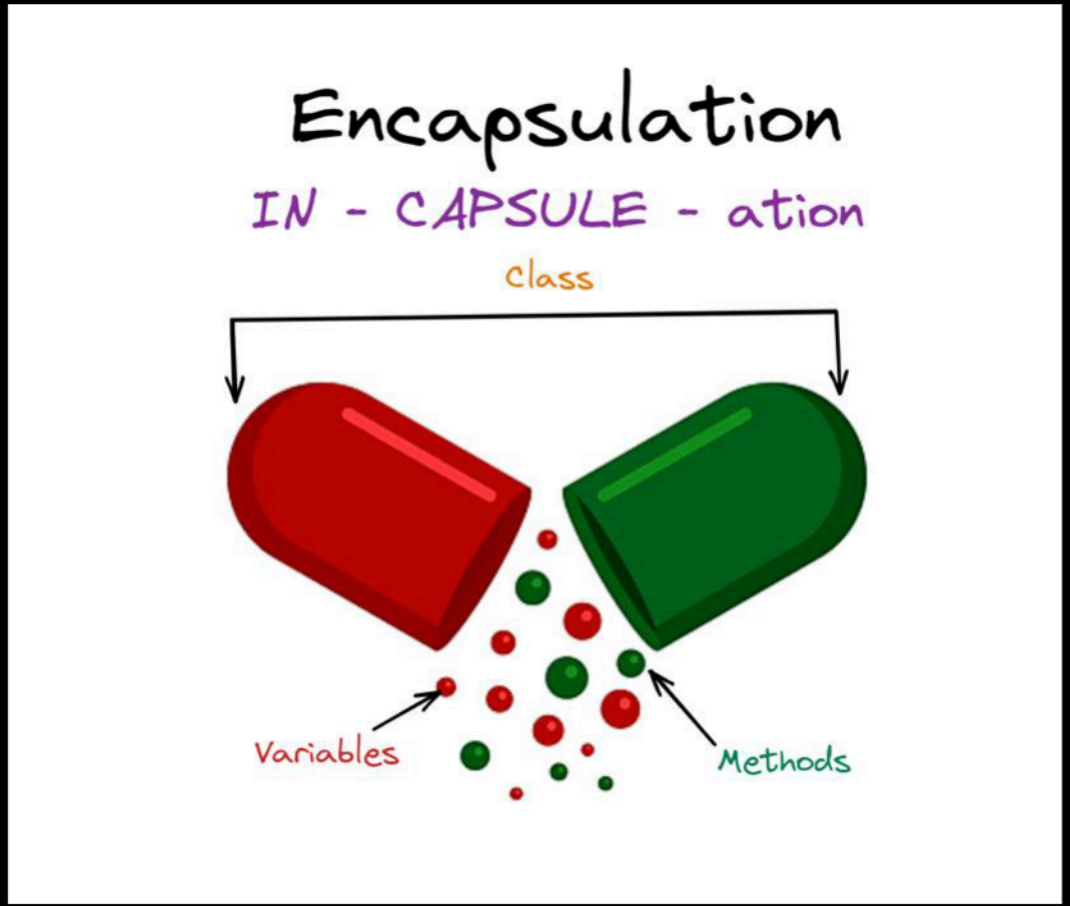


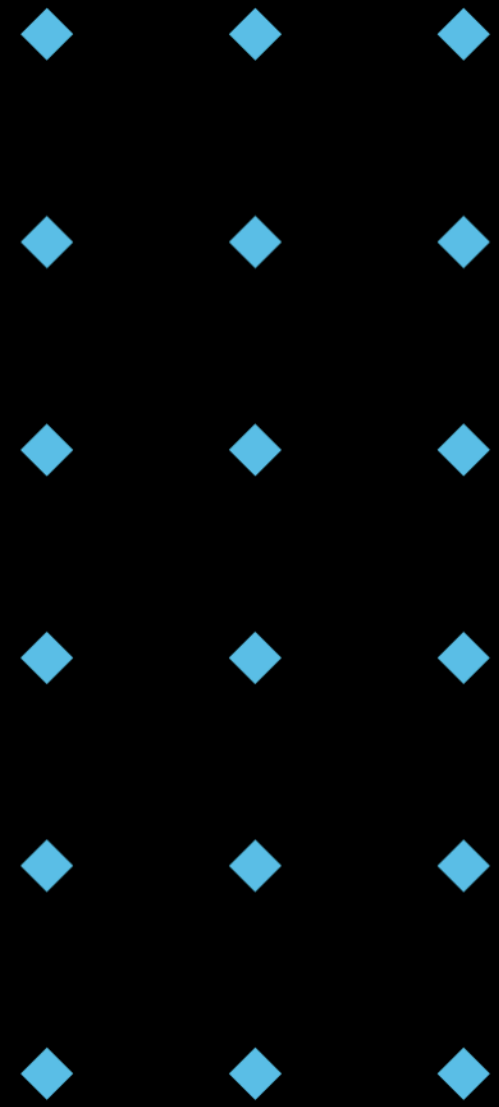
Image from AlgoDaily, year unknown,
https://algodaily.com/lessons/understanding-encapsulation-in-programming

Encapsulation: Core Principles

- **Data Hiding**: Protecting the internal state of an object.
- **Controlled Access**: Providing methods (getters and setters) to access and modify data.
- **Modularity**: Breaking down a system into independent, manageable units.
- **Abstraction**: Hiding complex implementation details and presenting a simplified interface.

Benefits of Encapsulation

- **Data Integrity**: Prevents accidental modification of data.
- **Enhanced Security**: Reduces the risk of unauthorized access.
- **Modularity**: Makes code easier to understand, test, and maintain.
- **Flexibility**: Allows internal implementation to change without affecting external code.
- **Reduced** Complexity: Simplifies the overall system by hiding unnecessary details.



Encapsulation in Action: Rectangle Class

```python
class Rectangle:
    def __init__(self, width, height):
        self.__width = width  # Private attribute
        self.__height = height # Private attribute

    def get_width(self):
        return self.__width

    def set_width(self, width):
        if width > 0:
            self.__width = width
        else:
            print("Width must be positive.")

    def get_height(self):
        return self.__height

    def set_height(self, height):
        if height > 0:
            self.__height = height
        else:
            print("Height must be positive.")

    def calculate_area(self):
        return self.__width * self.__height

    def display_dimensions(self):
        print(f"Width: {self.__width}, Height: {self.__height}")

rect = Rectangle(5, 10)
print(rect.get_width())  # Output: 5
rect.set_width(7)
print(rect.calculate_area())  # Output: 70
rect.display_dimensions()  # Output: Width: 7, Height: 10
```

Think-Pair-Share: Why Getters and Setters?

- **Question**: Why do we use getters and setters instead of directly accessing the width and height attributes in the Rectangle class?

* Take 2 minutes to think about it individually, then pair up with a classmate and discuss your answers for 3 minutes.

```python
class Rectangle:
    def __init__(self, width, height):
        self.__width = width  # Private attribute
        self.__height = height # Private attribute

    def get_width(self):
        return self.__width

    def set_width(self, width):
        if width > 0:
            self.__width = width
        else:
            print("Width must be positive.")

    def get_height(self):
        return self.__height

    def set_height(self, height):
        if height > 0:
            self.__height = height
        else:
            print("Height must be positive.")

    def calculate_area(self):
        return self.__width * self.__height

    def display_dimensions(self):
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rect = Rectangle(5, 10)
print(rect.get_width())  # Output: 5
rect.set_width(7)
print(rect.calculate_area())  # Output: 70
rect.display_dimensions()  # Output: Width: 7, Height: 10
```

Getters and Setters: Key Benefits

- **Data Validation**: Setters can validate input to ensure data integrity.
- **Controlled Access**: Allows read-only access or restricted modification.
- **Future Flexibility**: Internal implementation can change without affecting external code using the class.
- **Encapsulation**: Protects the internal state of the object.

Access Modifiers in Python

- Python doesn't have strict access modifiers like **private**, **protected**, and **public** found in other languages (e.g., Java, C++). Instead, it uses **naming conventions** to indicate the intended level of access.

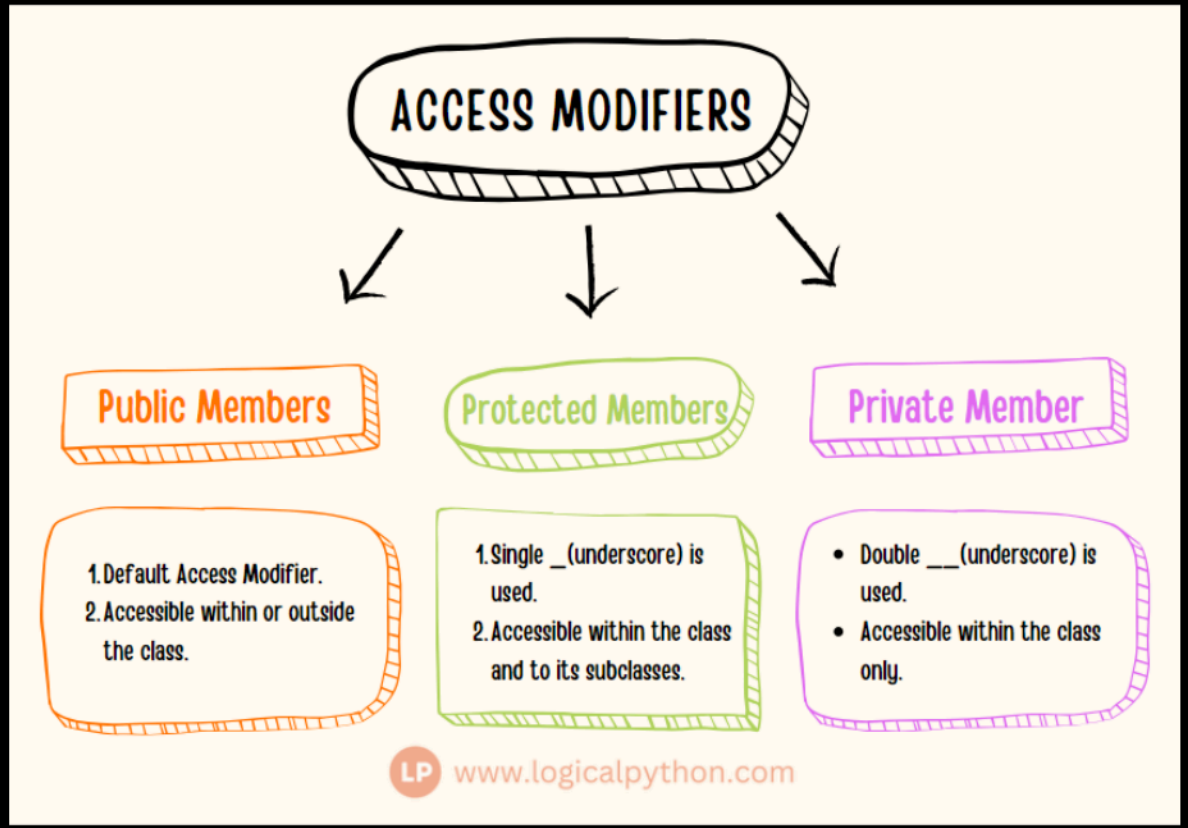

Image from Logicalpython, 2024, https://www.logicalpython.com/python-encapsulation-and-access-modifiers/

Public Access

- **Public** attributes and methods are accessible from anywhere – both inside and outside the class. No special naming convention is required.

```python
class Dog:
    def __init__(self, name):
        self.name = name  # Public attribute

    def bark(self): # Public method
        print("Woof!")


my_dog = Dog("Buddy")
print(my_dog.name) # Accessing public attribute
my_dog.bark() # Calling public method
```

Protected Access

- **Protected** attributes and methods are intended for internal use within the class and its subclasses. They are indicated by a **single underscore prefix _**.
- It's a signal to other developers that these should not be accessed directly from outside the class.

```python
class Vehicle:
    def __init__(self):
        self._engine_type = "Gasoline"  # Protected attribute

    def _start_engine(self): # Protected Method
        print("Engine Starting")

class Car(Vehicle):
    def display_engine_type(self):
        print(f"Engine type: {self._engine_type}")

my_car = Car()
my_car.display_engine_type()  # Accessing protected attribute from subclass
```

Private Access

- **Private** attributes and methods are intended for internal use only within the class. They are indicated by a **double underscore prefix** __.
- **Name Mangling**: Python uses name mangling to make these attributes harder (but not impossible) to access from outside the class.

```python
class BankAccount:
    def __init__(self, balance):
        self.__balance = balance  # Private attribute

    def deposit(self, amount):
        self.__balance += amount

    def get_balance(self):
        return self.__balance

account = BankAccount(100)
# print(account.__balance)  # This will raise an AttributeError
print(account._BankAccount__balance) # Accessing using name mangling, but discouraged
```

Name Mangling: How it Works

- When Python encounters an attribute or method name with a double underscore prefix, it changes the name to *_ClassName__AttributeName*.
- Purpose: To avoid accidental name collisions in subclasses and to provide a degree of privacy.
- Example: In the *BankAccount* class, *__balance* becomes *_BankAccount__balance*.

```
1   class BankAccount:
2       def __init__(self, balance):
3           self.__balance = balance  # Private attribute
4
5       def deposit(self, amount):
6           self.__balance += amount
7
8       def get_balance(self):
9           return self.__balance
10
11  account = BankAccount(100)
12  # print(account.__balance)  # This will raise an AttributeError
13  print(account._BankAccount__balance) # Accessing using name mangling, but discouraged
```

Benefits of Name Mangling

- **Reduces Name Collisions**: Avoids conflicts when subclasses have attributes with the same name as superclasses.
- **Encourages Good Design**: Signals to developers that an attribute is intended for internal use.
- **Provides a Degree of Privacy**: Makes it harder (but not impossible) to access private attributes from outside the class.

When to Use Name Mangling

- **Preventing Name Collisions**: When you want to avoid accidental name collisions in subclasses.
- **Indicating Internal Use**: To signal to developers that an attribute or method is intended for internal use only.
- **Protecting Sensitive Data**: To make it harder for external code to access sensitive data.

```python
class Parent:
    def __init__(self):
        self.__private_attribute = "Parent's Private"

    def get_private(self):
        return self.__private_attribute

class Child(Parent):
    def __init__(self):
        super().__init__()
        self.__private_attribute = "Child's Private"

    def get_child_private(self):
         return self.__private_attribute

parent = Parent()
child = Child()

print(parent.get_private())  # Output: Parent's Private
print(child.get_private())   # Output: Parent's Private (from Parent class)
print(child.get_child_private()) # Output: Child's Private
# Accessing mangled names
print(child._Parent__private_attribute) # Output: Parent's Private
print(child._Child__private_attribute) # Output: Child's Private

```

Name Mangling: Cautions

- **Not True Privacy**: It doesn't provide absolute privacy; attributes can still be accessed using name mangling.
- **Overuse**: Overusing name mangling can make code harder to read and understand.
- **Testability**: Can make unit testing more difficult if you rely heavily on accessing private attributes.

- Python's Philosophy: **"We are all responsible users**"
  - Python relies on the principle that developers are responsible and will respect the intended access levels.
  - Even private attributes can be accessed using name mangling.
  - It's a matter of convention and good practice, not strict enforcement.

https://docs.python-guide.org/writing/style/#we-are-all-responsible-users

Information Hiding: Beyond Encapsulation

- **Information hiding** is the principle of **concealing** the internal implementation details of a module or class from the outside world.
- To reduce **complexity**, improve **maintainability**, and increase **flexibility**.
- Encapsulation is a mechanism to achieve information hiding.

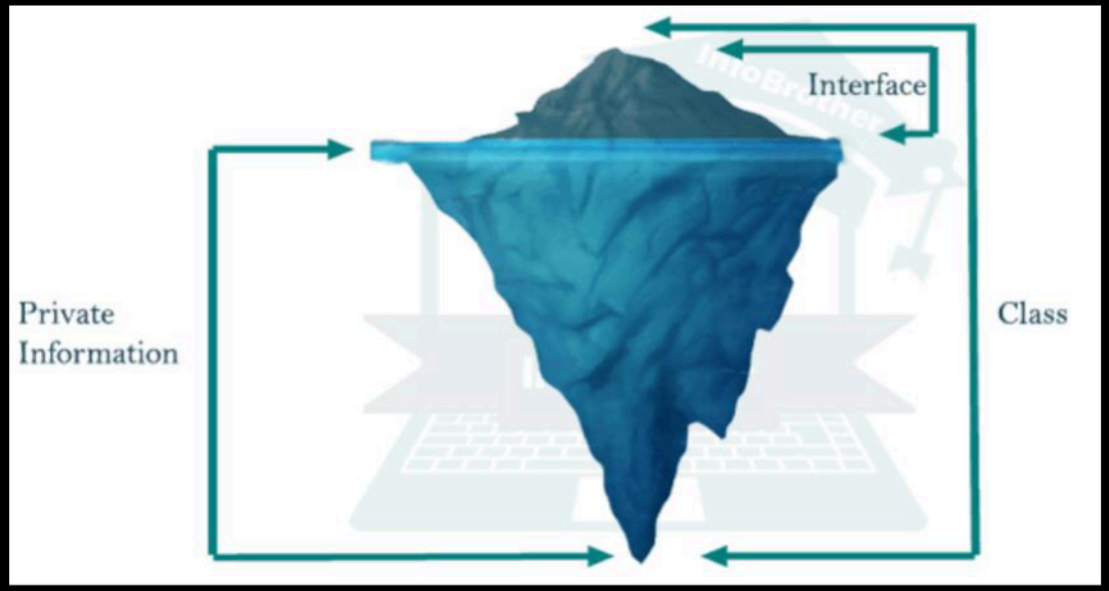

Image from Muntazir Fadhel, 2019, https://mfadhel.com/object-oriented-microservices-migrations/

Techniques for Information Hiding

- **Use Private Attributes**: Hide internal data using the __ prefix.
- **Implement Public Interfaces**: Provide well-defined methods for interacting with the class.
- **Abstract Complex Logic**: Hide complex calculations or algorithms behind simple method calls.
- **Use Helper Methods**: Break down complex methods into smaller, private helper methods.

Information Hiding Example: Temperature Converter

```python
class TemperatureConverter:
    def __init__(self):
        pass

    def celsius_to_fahrenheit(self, celsius):
        return (celsius * 9/5) + 32

    def fahrenheit_to_celsius(self, fahrenheit):
        return (fahrenheit - 32) * 5/9

    # Internal function that should be private
    def __kelvin_to_celsius(self, kelvin):
        return kelvin - 273.15

    def convert_to_standard(self, kelvin, standard="Celsius"):
        celsius = self.__kelvin_to_celsius(kelvin)
        if standard == "Fahrenheit":
            return self.celsius_to_fahrenheit(celsius)
        return celsius

converter = TemperatureConverter()
print(converter.celsius_to_fahrenheit(25))  # Output: 77.0
print(converter.fahrenheit_to_celsius(77))  # Output: 25.0
```

Benefits of Information Hiding

- **Reduced Complexity**: Simplifies the interface for users of the class.
- **Increased Maintainability**: Allows internal implementation to be changed without affecting external code.
- **Improved Security**: Prevents unauthorized access to sensitive data.
- **Enhanced Flexibility**: Makes it easier to adapt the class to new requirements.

Getters and Setters: Controlling Access

- **Getters** (also known as accessors) are methods used to retrieve the value of an attribute.
- **Setters** (also known as mutators) are methods used to modify the value of an attribute.
- Purpose: To provide controlled access to an object's attributes and to enforce data validation.

- OOP and encapsulation are not remedies for bad architecture! If all your classes need access to each others private variables, the problem is not OOP.

Getter and Setter Example

- The *get_name()* method provides read-only access to the *__name* attribute.
- The *set_name()* method validates the input before modifying the *__name* attribute.

```python
class Person:
    def __init__(self, name):
        self.__name = name

    def set_name(self, new_name):
        if isinstance(new_name, str) and len(new_name) > 0:
            self.__name = new_name
        else:
            print("Invalid name.")

    def get_name(self):
        return self.__name

person = Person("Alice")
person.set_name("Bob")
print(person.get_name())  # Output: Bob
```

Pythonic Getters and Setters: Properties

- Python provides a more elegant way to define getters and setters using the *@property* decorator.
- The *@property* decorator defines the getter, and the *@radius.setter* decorator defines the setter.
- Question: How to use *@property* to stop others from accessing the radius?

```python
class Circle:
    def __init__(self, radius):
        self.__radius = radius

    @property
    def radius(self):
        return self.__radius

    @radius.setter
    def radius(self, value):
        if value > 0:
            self.__radius = value
        else:
            print("Radius must be positive.")

circle = Circle(5)
print(circle.radius)  # Output: 5
circle.radius = 7
print(circle.radius)  # Output: 7
```

Think-Pair-Share: Why Properties?

- Question: What are the advantages of using properties compared to traditional getter and setter methods?

```python
class Circle:
    def __init__(self, radius):
        self.__radius = radius

    @property
    def radius(self):
        return self.__radius

    @radius.setter
    def radius(self, value):
        if value > 0:
            self.__radius = value
        else:
            print("Radius must be positive.")

circle = Circle(5)
print(circle.radius)  # Output: 5
circle.radius = 7
print(circle.radius)  # Output: 7
```

Properties: Advantages

- **Readability**: Makes code more concise and easier to understand.
- **Maintainability**: Simplifies the process of adding validation or other logic to attribute access.
- **Flexibility**: Allows you to change the implementation of attribute access without affecting external code.

Encapsulation Best Practices

- Use **Access Modifiers** (Conventions): Follow the public, protected, and private naming conventions.
- Use **Getters** and **Setters** (Properties): Provide controlled access to attributes.
- **Hide Implementation Details**: Conceal complex logic and data structures.
- Favor **Immutability**: When possible, make objects immutable to prevent accidental modification.
- **Document** Your Code: Clearly document the intended use of attributes and methods.

Immutability: A Powerful Tool

- Immutable objects **cannot** be modified after they are created.
- Benefits:
  - **Simplified Reasoning**: Easier to understand and reason about the behaviour of immutable objects.
  - **Thread Safety**: Immutable objects are inherently thread-safe.
  - **Reduced Errors**: Prevents accidental modification of data.

```python
from collections import namedtuple

Point = namedtuple("Point", ["x", "y"])
p1 = Point(1, 2)
# p1.x = 3  # This will raise an AttributeError because Point is immutable
p2 = Point(3,4)
```

The Importance of Documentation

- **Clear and concise documentation** is crucial for understanding and maintaining encapsulated code.
- What to Document:
  - Purpose of attributes and methods.
  - Intended access levels (public, protected, private).
  - Any validation rules or constraints.
  - Any potential side effects.

Refactoring for Better Encapsulation

- Scenario: You have existing code that doesn't follow encapsulation principles.
- Steps:
  - Identify attributes that should be **private**.
  - Create **getters** and **setters** (properties) for controlled access.
  - Move complex logic into **private helper methods**.
  - Add **documentation** to explain the intended use of attributes and methods.

Encapsulation: Common Pitfalls

- **Over-Encapsulation**: Making everything private can lead to unnecessary complexity.
- **Under-Encapsulation**: Exposing too much internal state can lead to data corruption and maintenance problems.
- **Ignoring Conventions**: Not following the public, protected, and private naming conventions.
- **Not Documenting Your Code**: Makes it difficult for others (and yourself!) to understand and maintain the code.

Encapsulation: Inside Your Car

- Designing a Car class to represent your vehicle.

- *model* and *color*: Public. These are external characteristics readily observable and known to anyone who sees the car. It's the overall appearance!
- *__engine_running* and *__fuel_level*: Private. These represent the internal state of the car's mechanics. You don't directly manipulate these from outside, but control them through methods like *start_engine()*, *stop_engine()*, and *refuel()*.

```python
class Car:
    def __init__(self, model, color):
        self.model = model  # Public: Everyone knows what model it is
        self.color = color  # Public: Everyone can see the colour
        self.__engine_running = False  # Private: Start the engine with the key
        self.__fuel_level = 100  # Private: Internal fuel level

    def start_engine(self, car_key):
        if car_key == "car_key_number":
            self.__engine_running = True
            print("Vroom! Engine started.")

    def stop_engine(self):
        self.__engine_running = False
        print("Silence. Engine stopped.")

    def drive(self, distance):
        if self.__engine_running:
            if self.__fuel_level > 0:
                print(f"Driving {distance} miles.")
                self.__fuel_level -= distance / 10  # Simulating fuel consumption
                if self.__fuel_level < 0:
                    self.__fuel_level = 0
                print(f"Fuel level: {self.__fuel_level}%")
            else:
                print("Out of fuel!  Need to refuel.")
        else:
            print("Engine is not running. Start the engine first.")

    def refuel(self, amount):
        self.__fuel_level += amount
        if self.__fuel_level > 100:
            self.__fuel_level = 100
        print(f"Refueled. Fuel level: {self.__fuel_level}%")


# Example Usage
my_car = Car("Tesla Model 3", "Red")
print(f"My car is a {my_car.color} {my_car.model}") #publicly shown
my_car.start_engine() #start engines
my_car.drive(50) #drive
my_car.stop_engine() #stop the engine
```

Key Takeaways

- Encapsulation is bundling and hiding data.
- Access modifiers in Python are convention-based (public, protected, private).
- Information hiding reduces complexity and improves maintainability.
- Getters and setters (properties) provide controlled access to attributes.
- Name mangling helps prevent name collisions and provides a degree of privacy.
- Following best practices is crucial for writing well-encapsulated code.

Q & A

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Lab